Pilda 1

$CPI_{ideal} = 8,5$ CC

Memory accesses per instruction = 3

Miss Rate = 11%

Miss Penalty = 6 CC

Clock cycle time = 4 ns

AMAT = ?

CPU_{TIME} = ?

$$AMAT = (1 + MR \cdot \overset{CC}{MP}) \cdot CCT = (1 + 0,11 \cdot 6) \cdot 4 = 1,66 \cdot 4 = 6,64 \text{ ns}$$

$$CPU_{time} = ic \cdot (CPI_{ideal} + MR \cdot \overset{CC}{MP} \cdot \text{Mem accesses per instr}) \cdot CCT =$$

$$= ic \cdot (8,5 + 0,11 \cdot 6 \cdot 3) \cdot 4 = 41,92 \cdot ic \text{ ns}$$

$$\frac{\text{Performance cache}}{\text{Performance ideal}} = \frac{CPU_{time\ ideal}}{CPU_{time\ real}} = \frac{\cancel{ic} \cdot CPI_{ideal} \cdot CCT}{41,92 \cdot \cancel{ic}} = \frac{34}{41,92} = 0,811$$

$\Rightarrow$ performanța cache-ului e cu 0,189 mai slabă

Pilda 2

64 KiB cache data

CCT = 20 ns

Mem ac / instr. = 1,3

$CPI_{ideal} = 1,5$ CC

Pt acest cache, dacă se adopta SA, det. o degradare a frecvenței clock-ului cu 8,5%. În acest context, se analizează cache-ul în felul urm: DM vs. 2 WAY SA

	DM	2 WAY SA
Miss Rate	3,9%	3%

Miss Penalty = 200 ns

$$AMAT_{BM} = (1 + MR \cdot \overset{\nearrow CC}{MP}) \cdot CCT = (1 + 0{,}039 \cdot 200) \cdot 20 = 27{,}8\ ms$$

$$AMAT_{SA} = (1 + 0{,}03 \cdot 200) \cdot 21{,}7 = 27{,}7\ ms$$

$$CCT_{SA} = 1{,}085 \cdot CCT_{BM} = 21{,}7\ ms$$

$\Rightarrow$ $AMAT_{SA}$ is better than $AMAT_{BM}$

$$CPU_{time_{BM}} = ic \cdot (CPi_{ideal} + MR \cdot \overset{\nearrow CC}{MP} \cdot Mem.\ ac./instr) \cdot CCT =$$

$$= ic \cdot (CPi_{ideal} \cdot CCT + MR \cdot \overset{\nearrow t}{MP} \cdot Mem\ ac/instr) =$$

$$= ic\,(1{,}5 \cdot 20 + 0{,}039 \cdot 200 \cdot 1{,}3) =$$

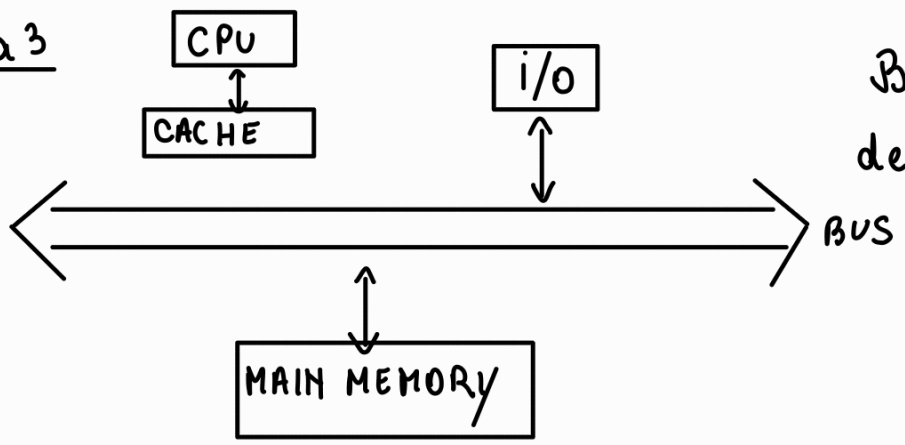
$$= ic \cdot 40{,}14\ ms$$

$$CPU_{time_{SA}} = ic\,(1{,}5 \cdot 21{,}7 + 0{,}03 \cdot 200 \cdot 1{,}3) =$$

$$= ic \cdot 40{,}35\ ms$$

$\Rightarrow$ BM mai rapid decât SA

Pilda 3



Cât la % din BUS Bandwidth este „mâncat" de imperfecțiunea cache-ului?

BUS bandwidth $= 10^9$ words/sec

BUS width = 2 words

Hit Rate = 90% $\longrightarrow$ MR = 10%

1 block = 4 words

CPU reference frequency $= 10^8$ words/sec

Pp. că în (∀) mom 35% din blocurile cache-ului au fost modificate (sunt dirty). Cache are o politică W.A., frecvența scrierilor e de 30%

a) WB b) WT

a) Write Miss Penalty = Read Miss Penalty

Accesses via the BUS = $\underset{freq}{10^8} \cdot \underset{MR}{0,1} \cdot (0,3 \cdot WMP^b_{WB} + (1-0,3)\, RMP^b_{WB})$

$$RMP^b_{WB} = \underbrace{\frac{\text{block size}}{\text{bus width}} \cdot (\text{Bus Reads})}_{\text{Allocate}} + \underbrace{\text{dirty} \cdot \frac{\text{block size}}{\text{bus width}} (\text{Bus Writes})}_{\text{Update}} =$$

$$= \frac{4}{2} + 0,35 \cdot \frac{4}{2} = 2,7 \text{ BUS accesses} = WMP^b_{WB}$$

Accesses via the BUS $= 10^7 \cdot (0,3 \cdot 2,7 + 0,7 \cdot 2,7) = 10^7 \cdot 2,7$

$$\Rightarrow \%\text{ Bus Used}_{WB} = \frac{\text{Acc. via the BUS}}{\text{BUS bandwidth}} = \frac{10^7 \cdot 2,7}{10^9} = \frac{2,7}{100} = 2,7\%$$

b) WT

Acc via the BUS $= 10^8 \cdot 0,1 \underbrace{(0,3 \cdot WMP^b_{WT} + 0,7 \cdot RMP^b_{WT})}_{\text{Miss}} + \underbrace{10^8 \cdot \overset{HR}{0,9} \cdot WHP^b_{WT} \cdot \underset{freq\ rate}{0,3}}_{\text{Hit P}}$

$WMP^b_{WT} = \underbrace{\frac{4}{2} \text{ Bus Reads}}_{ALL} + \underset{MM}{1 \text{ Bus Write}} = 3 \text{ BUS accesses}$

$RMP^b_{WT} = \frac{4}{2} \text{ BUS Reads} = 2 \text{ BUS accesses}$

$WHP^b_{WT} = 1 \text{ BUS Write} = 1 \text{ BUS accesses}$

acc. via the BUS $= 10^7 \cdot (0,3 \cdot 3 + 0,7 \cdot 2) + 9 \cdot 10^7 \cdot 1 \cdot 0,3 = 5 \cdot 10^7$

$$\Rightarrow \%\text{ BUS width} = \frac{5 \cdot 10^7}{10^9} = 5\%$$